

Algunas Propiedades

- $J_n(x) = J_{-n}(x)$
- Usando la función generadora $E(x, z)$ es posible probar que las funciones de Bessel $J_n(x)$ satisfacen las relaciones de recurrencia

$$(a) \quad \frac{2n}{x} J_n(x) = J_{n-1}(x) + J_{n+1}(x)$$

$$(b) \quad 2 J'_n(x) = J_{n-1}(x) - J_{n+1}(x)$$

$$(c) \quad J_{-1}(x) = -J_1(x); \quad J'_0(x) = -J_1(x)$$

$$(d) \quad \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$$

$$(e) \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x)$$

- Se definen genéricamente como funciones de Bessel aquellas funciones $Z_\nu(x)$ que satisfacen

$$(a) \quad \frac{2\nu}{x} Z_\nu(x) = Z_{\nu-1}(x) + Z_{\nu+1}(x)$$

$$(b) \quad 2 Z'_\nu(x) = Z_{\nu-1}(x) - Z_{\nu+1}(x)$$

$$x^2 \frac{d^2 Z_\nu(x)}{dx^2} + x \frac{d Z_\nu(x)}{dx} + (x^2 - \nu^2) Z_\nu(x) = 0$$

REPRESENTACIÓN INTEGRAL DE LAS FUNCIONES de Bessel

TEOREMA: Si n es un entero no negativo, entonces
($n = n$ entero no negativo)

$$J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta \quad (1)$$

Resultado conocido como Representación integral de las Funciones de Bessel.

Prueba: Dado que $G(x,t) = e^{\frac{x}{2}(t - \frac{1}{t})}$,

$$\Rightarrow t - \frac{1}{t} = e^{i\theta} - e^{-i\theta}$$

de modo que $G(x,t) = e^{\frac{x}{2}(t - \frac{1}{t})} = e^{\frac{x}{2} \cdot 2i \sin \theta}$

$$G(x,t) = e^{ix \sin \theta} \quad (2)$$

Además,

$$G(x,t) = e^{\frac{x}{2}(t - \frac{1}{t})} = \sum_{n=-\infty}^{\infty} J_n(x) t^n = \sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta}$$

$$\Rightarrow e^{ix \sin \theta} = \sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta} \quad (3)$$

$$\sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta} = J_0 + J_1 e^{i\theta} + J_{-1} e^{-i\theta} + J_2 e^{i2\theta} + J_{-2} e^{-i2\theta} + \dots$$

Puesto que $J_n(x) = (-1)^n J_{-n}(x)$, tenemos

$$\begin{aligned} \sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta} &= J_0 + J_1 (e^{i\theta} - e^{-i\theta}) + J_2 (e^{i2\theta} + e^{-i2\theta}) \\ &\quad + J_3 (e^{i3\theta} - e^{-i3\theta}) + J_4 (e^{i4\theta} + e^{-i4\theta}) + \dots \end{aligned}$$

$$\begin{aligned} \text{Pero, } e^{i\theta} - e^{-i\theta} &= 2i \sin \theta \\ e^{i2\theta} + e^{-i2\theta} &= 2 \cos 2\theta, \text{ etc} \end{aligned}$$

$$\sum_{n=-\infty}^{\infty} J_n(x) e^{in\theta} = J_0 + i2 J_1 \sin \theta + 2 J_2 \cos 2\theta \\ + i2 J_3 \sin 3\theta + 2 J_4 \cos 4\theta + \dots$$

$$= J_0 + 2 [J_2 \cos 2\theta + J_4 \cos 4\theta + J_6 \cos 6\theta + \dots] \\ + i2 [J_1 \sin \theta + J_3 \sin 3\theta + J_5 \sin 5\theta + \dots]$$

Así

$$\cos(x \sin \theta) = J_0 + 2 [J_2 \cos 2\theta + J_4 \cos 4\theta + \dots J_{2n} \cos 2n\theta]$$

$$\sin(x \sin \theta) = 2 [J_1 \sin \theta + J_3 \sin 3\theta + \dots + J_{2n-1} \sin(2n-1)\theta]$$

o bien

$$\cos(x \sin \theta) = J_0 + 2 \sum_{n=1}^{\infty} J_{2n}(x) \cos(2n\theta)$$

$$\sin(x \sin \theta) = 2 \sum_{n=1}^{\infty} J_{2n-1}(x) \sin(2n-1)\theta$$

De la ortogonalidad de las Funciones Seno y Coseno Sabemos,

$$\int_0^{\pi} \cos n\theta \sin m\theta d\theta = 0$$

$$\int_0^{\pi} \cos n\theta \cos m\theta d\theta = \frac{\pi}{2} \delta_{nm}$$

$$\int_0^{\pi} \sin n\theta \sin m\theta d\theta = \frac{\pi}{2} \delta_{nm}$$

Multiplícamo por $\cos n\theta$ y $\sin n\theta$ respectivamente e integrando tenemos:

$$\bullet \int_0^\pi \cos m\theta \cos(x \sin \theta) d\theta = \int_0^\pi J_0 \cos m\theta d\theta$$

$$+ 2 \sum_{n=1}^{\infty} J_{2n}(x) \int_0^\pi \cos m\theta \cos(2n\theta) d\theta$$

$$\bullet \int_0^\pi \sin m\theta \sin(x \sin \theta) d\theta = 2 \sum_{n=1}^{\infty} J_{2n-1}(x) \int_0^\pi \sin m\theta \sin(2n-1)\theta d\theta$$

Dado que

$$\int_0^\pi \cos m\theta d\theta = \frac{1}{m} \sin m\theta \Big|_0^\pi = 0$$

$$\int_0^\pi \cos m\theta \cos(2n\theta) d\theta = \frac{\pi}{2} \delta_{m,2n}$$

$$\Rightarrow \int_0^\pi \cos m\theta \cos(x \sin \theta) d\theta = 0 + 2 \sum_{n=1}^{\infty} \underline{J_{2n}(x)} \frac{\pi}{2} \underline{\delta_{m,2n}}$$

$$= \pi J_m, \quad m \text{ PAR}$$

$$\Rightarrow \frac{1}{\pi} \int_0^\pi \cos m\theta \cos(x \sin \theta) d\theta = \begin{cases} J_m(x), & m \text{ PAR} \\ 0, & m \text{ IMPAR} \end{cases}$$

$$\text{Dado que } \int_0^\pi \sin m\theta \sin(2n-1)\theta d\theta = \frac{\pi}{2} \delta_{m,2n-1}$$

$$\Rightarrow \int_0^\pi \sin m\theta \sin(x \sin \theta) d\theta = 2 \sum_{n=1}^{\infty} \underline{J_{2n-1}(x)} \frac{\pi}{2} \underline{\delta_{m,2n-1}}$$

$$= \pi J_m, \quad m \text{ IMPAR}$$

$$\Rightarrow \frac{1}{\pi} \int_0^\pi \sin m\theta \sin(x \sin \theta) d\theta = \begin{cases} 0, & \text{si es PAR} \\ J_m, & \text{si } m \text{ es IMPAR} \end{cases}$$

Sumando,

$$\frac{1}{\pi} \int_0^{\pi} \cos m\theta \cos(x \sin \theta) d\theta + \frac{1}{\pi} \int_0^{\pi} \sin \theta \sin(x \sin \theta) d\theta$$

$$= \begin{cases} J_m, m \text{ Par} \\ J_m, m \text{ impar} \end{cases}$$

$$\Rightarrow \frac{1}{\pi} \int_0^{\pi} \cos(m\theta - x \sin \theta) d\theta = J_m(x)$$

Así tenemos

$$J_n(x) = \frac{1}{\pi} \int_0^{\pi} \cos(n\theta - x \sin \theta) d\theta, n = 0, 1, 2, 3, \dots$$

Caso especial: $n = 0$

$$J_0(x) = \frac{1}{\pi} \int_0^{\pi} \cos(x \sin \theta) d\theta$$

ORTOGONALIDAD de las Funciones de Bessel.

En coordenadas cilíndricas (ρ, φ, z) , la ec. de Helmholtz es dada por:

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 \psi}{\partial \varphi^2} + \frac{\partial^2 \psi}{\partial z^2} + k^2 \psi = 0 \quad (1)$$

cuya solución es dada por

$$\psi(\rho, \varphi, z) = P(\rho) \Phi(\varphi) Z(z) \quad (2)$$

Sustituyendo (2) en (1) obtenemos

$$\frac{d^2 Z}{dz^2} + l^2 Z = 0 \quad ; \quad \frac{d^2 \Phi}{d\varphi^2} + m^2 \Phi = 0$$

$$\rho \frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + (n^2 \rho^2 - \lambda_2) P = 0 \quad (3)$$

$\lambda_2 = -m^2$ ($m = 0, 1, 2, \dots$) de modo que la ec (3)

$$\frac{d}{d\rho} \left(\rho \frac{dP}{d\rho} \right) + \left(n^2 \rho - \frac{m^2}{\rho} \right) P = 0 \quad (4)$$

Si imponemos condiciones de borde sobre $P(\rho)$
 $\Rightarrow$ la ec (4) se convierte en una ec. de eigenvalores con eigenvalor n^2 .

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] - s(x)y + \lambda r(x)y$$

Algunas de las condiciones Físicas más comunes

(a) en $\rho = 0$, $P(\rho)$ debe ser finita

(b) en $\rho = a$, $P(\rho)$ debe satisfacer

$P(a) = 0$: condición de Dirichlet o -

$\left. \frac{dP}{d\rho} \right|_{\rho=a} = 0$: condición de Neumann o -

$A\rho + B \left. \frac{dP}{d\rho} \right|_{\rho=a} = 0$; condición de borde intermedio.

— o —

Cons. LA EDO de Bessel

$$x^2 \frac{d^2 Z_\nu(x)}{dx^2} + x \frac{dZ_\nu(x)}{dx} + (x^2 - \nu^2) Z_\nu(x) = 0$$

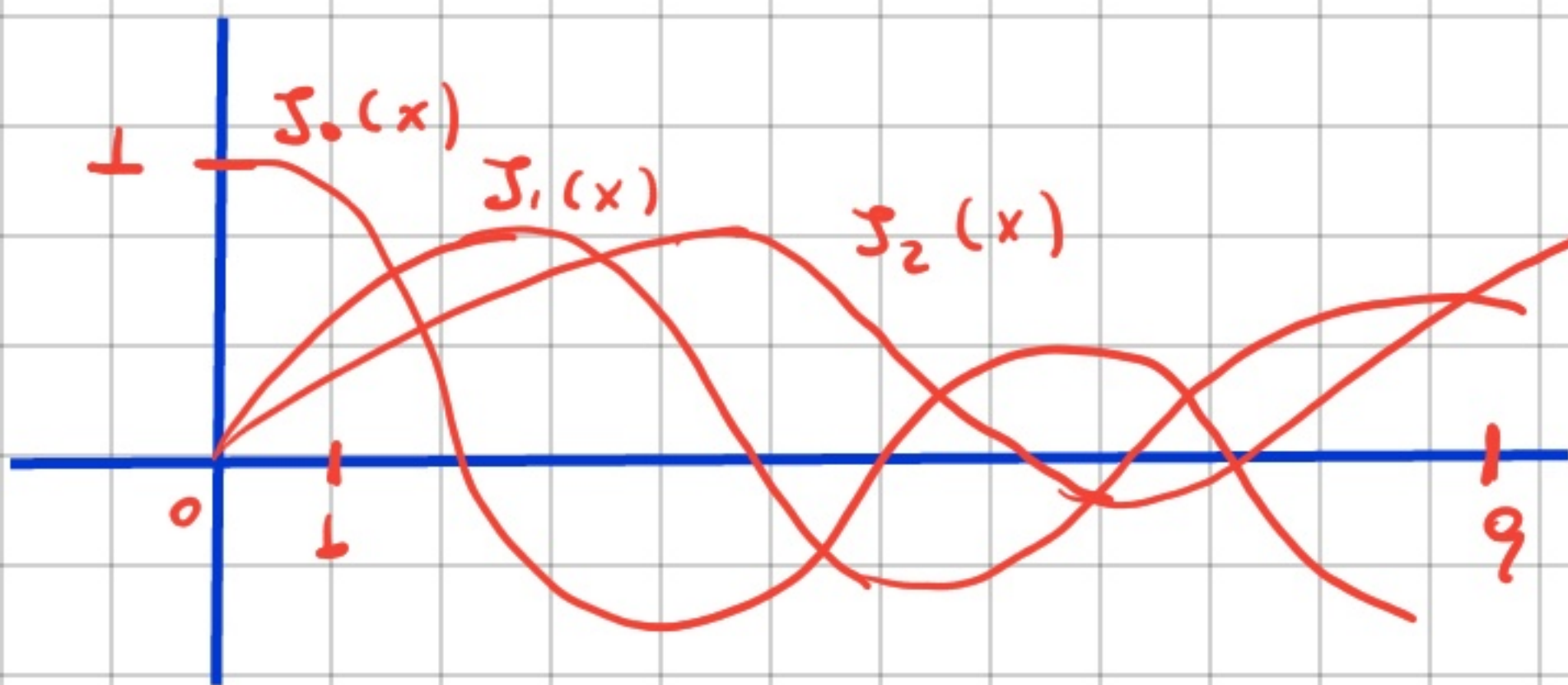
Por ejemplo de 0 hasta a ; $x \rightarrow k\rho$

$$\cancel{k^2} p^2 \frac{d^2 Z_\nu(kp)}{\cancel{k^2} dp^2} + \cancel{k} p \frac{d Z_\nu(kp)}{\cancel{k} dp} + (k^2 p^2 - \omega^2) Z_\nu(kp) = 0$$

Puede ser escrita como

$$\frac{d}{dp} \left(p \frac{d Z_\nu(kp)}{dp} \right) + \left(k^2 p - \frac{\omega^2}{p} \right) Z_\nu(kp) = 0$$

$Z_\nu(kp)$ y $Z_{-\nu}(kp)$ } soluciones



Denotaremos las raíces de $J_\nu(x)$ por $\alpha_{\nu m}$, esto significa que

$$J(kp) \Big|_{kp = \alpha_{\nu m}} = 0, \quad J(\alpha_{\nu m}) = 0$$

Por otro lado la condición de Dirichlet

$$J(kp) \Big|_{p=a} = 0$$

de modo que los valores apropiados para k , considerando las condiciones de Dirichlet,

$$k_{\nu m} a = \alpha_{\nu m} \Rightarrow k_{\nu m} = \frac{\alpha_{\nu m}}{a}$$

Así, ($J_\nu \equiv Z_\nu$)

$$\frac{d}{dp} \left(p \frac{d J_\nu \left(\frac{\alpha_{\nu m}}{a} p \right)}{dp} \right) + \left(\frac{\alpha_{\nu m}^2 p}{a^2} - \frac{\omega^2}{p} \right) J_\nu \left(\frac{\alpha_{\nu m}}{a} p \right) = 0$$

Consideremos la ec. anterior para α_{lm}

$$\frac{d}{d\rho} \left(\rho \frac{dJ_l}{d\rho} \left(\frac{\alpha_{lm}}{a} \rho \right) \right) + \left(\frac{\alpha_{lm}^2 \rho}{a^2} - \frac{l^2}{\rho} \right) J_l \left(\frac{\alpha_{lm}}{a} \rho \right) = 0$$

Multiplicando por $J_l \left(\frac{\alpha_{lm}}{a} \rho \right)$ y por $J_l \left(\frac{\alpha_{lm}}{a} \rho \right)$

Respectivamente,

$$\begin{aligned} \cdot J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \frac{d}{d\rho} \left\{ \rho \frac{d}{d\rho} J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \right\} + \left(\frac{\alpha_{lm}^2 \rho}{a^2} - \frac{l^2}{\rho} \right) \\ \times J_l \left(\alpha_{lm} \frac{\rho}{a} \right) J_l \left(\alpha_{lm} \frac{\rho}{a} \right) = 0 \end{aligned}$$

$$\begin{aligned} \cdot J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \frac{d}{d\rho} \left\{ \rho \frac{d}{d\rho} J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \right\} + \left(\frac{\alpha_{lm}^2 \rho}{a^2} - \frac{l^2}{\rho} \right) \\ \times J_l \left(\alpha_{lm} \frac{\rho}{a} \right) J_l \left(\alpha_{lm} \frac{\rho}{a} \right) = 0 \end{aligned}$$

Restando,

$$\begin{aligned} J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \frac{d}{d\rho} \left[\rho \frac{d}{d\rho} J_l \left(\alpha_{lm} \frac{\rho}{a} \right) - J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \frac{d}{d\rho} J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \right] \\ + \frac{\alpha_{lm}^2 - \alpha_{lm}^2}{a^2} \rho J_l \left(\alpha_{lm} \frac{\rho}{a} \right) J_l \left(\alpha_{lm} \frac{\rho}{a} \right) = 0 \end{aligned}$$

Ecuación que puede ser re-escrita de la forma

$$\begin{aligned} \frac{d}{d\rho} \left[\rho J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \frac{d}{d\rho} J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \right] - \frac{d}{d\rho} \left[\rho J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \right. \\ \left. \times \frac{d}{d\rho} J_l \left(\alpha_{lm} \frac{\rho}{a} \right) \right] \end{aligned}$$

$$- \int \frac{d}{df} J_n(\alpha_{nm} f/a) \frac{d}{df} J_m(\alpha_{nm} f/a)$$

$$+ \int \frac{d}{df} J_n(\alpha_{nm} f/a) \frac{d}{df} J_n(\alpha_{nm} f/a)$$

$$= \frac{\alpha_{nn}^2 - \alpha_{nm}^2}{a^2} \int J_n(\alpha_{nm} \frac{f}{a}) J_n(\alpha_{nm} \frac{f}{a})$$

Integrando entre 0 y a tenemos

$$\int_0^a \frac{d}{df} \left\{ \int J_n(\alpha_{nm} \frac{f}{a}) \frac{d}{df} J_n(\alpha_{nm} \frac{f}{a}) - \int J_n(\alpha_{nm} \frac{f}{a}) \right. \\ \left. \times \frac{d}{df} J_n(\alpha_{nm} \frac{f}{a}) \right\} df$$

$$= \frac{\alpha_{nn}^2 - \alpha_{nm}^2}{a} \int_0^a \int J_n(\alpha_{nm} f/a) J_n(\alpha_{nm} f/a) df$$

$$\Rightarrow \int J_n(\alpha_{nm} f/a) \frac{d}{df} J_n(\alpha_{nm} f/a) \Big|_{f=0}^{f=a}$$

$$- \int J_n(\alpha_{nm} f/a) \frac{d}{df} J_n(\alpha_{nm} f/a) \Big|_{f=0}^{f=a}$$

$$= \frac{\alpha_{nn}^2 - \alpha_{nm}^2}{a} \int_0^a \int J_n(\alpha_{nm} \frac{f}{a}) J_n(\alpha_{nm} \frac{f}{a}) df$$

$\Rightarrow$ Para $n \neq m$ se tiene

$$\int_0^a \int J_n(\alpha_{nm} \frac{f}{a}) J_n(\alpha_{nm} \frac{f}{a}) df = 0 \quad //$$

$$\bullet \int_0^a \left[J_n(\alpha_{nm} \frac{f}{a}) \right]^2 f df = \frac{a^2}{2} [J_{n+1}(\alpha_{nm})]^2$$